3/5 Exp- 1

a) why freezing point depression occur when a non-volatile and non electrolyte solute is added to a solution

The freezing point depression occure when a non volatile and non electrolyte solute is added to the solution ~~becco~~ because of lowering of vapour pressure, that is directly proportional to the depression of freezing b point by the following expression

$$\Delta P \propto \Delta T_f .$$

b) when ΔT_f will be eq. l to K_f ?

we know that $\Delta T_f = K_f m$ ——— (1)

where, ΔT_f = freezing point depression

K_f = cryscopic constant

m = molality

when $m = 1$, for 1 molal solution we get from equation (1) $\Delta T_f = K_f$.

Define colligative properties. Why colligative properties of electrolytes and non electrolytes are not same?

colligative properties: Those properties of the solutions which depend only upon the ~~the~~ total number of molecules of solute per unit volume and not on it's chemical nature are called colligative properties.

The name of colligative properties are given below

i) vapour pressure lowering
ii) boiling point elevation
iii) freezing point depression
iv) osmotic pressure

colligative properties of electrolytes and non electrolytes are not same because electrolytes undergo dissociation or association in solution.

For dissociation of electrolytes the number of solute particles in solution is increased, therefore the colligative properties are increased. For association of colligative properties, the number of solute particles in solution are decreased so, colligative properties decreased. For non electrolyte there is no dissociation or association occur.

what is freezing point depression? write equation showing relation between molecular mass of a non-volatile and non electrolyte solute and depression of freezing point. will electrolyte and non-electrolyte depress freezing point equally? Explain the reason.

Freezing point: of a solution is the temp at which the solution is in equilibrium with the pure crystalline solvent

Freezing point depression.

$$\Delta T_f = T_f^{\circ} - T_f$$

T_f → freezing point temp of solution

T_f° → Freezing point temp of pure solvent

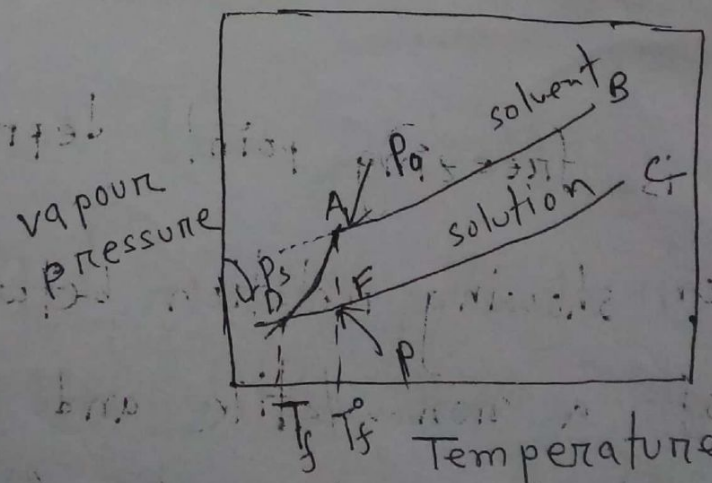


Fig: Depression of freezing point

In Fig vapour pressure curve of liquid and solvent and solid solvent meet at A. Hence this corresponds to the freezing point T_f° of solvent o and T_f is the freezing point of the solution. T_f is lower than T_f°.

A quantitative relationship between the concentration of solute in solution and depression of freezing point was given by F-M. Raoult

This relationship derived easily with the help of of above Figure.

Let p° be the vapour pressure of the solvent at its freezing point T_f°, and p and p_s are the vapour pressure of the solution at T_f° and T_f its freezing point respectively,

For very dilute solution, the freezing point depression is very small and the line segments AF and DF may be considered as straight line.

vapour pressure curve for a number of solution near the freezing point will be parallel then $\frac{DF}{AF}$ will be constant

i.e $\frac{DF}{AF} = K$

But the length DF equal to ΔT_f, AF is proportional to $(p^0 - p)$. If K is replaced by new constant $K' (= K.p^0)$ then equation becomes

$$\frac{\Delta T_f}{p^0 - p} = \frac{K'}{p^0}$$

or $\Delta T_f = K' \frac{(p^0 - p)}{p^0}$

p^0, the vapour pressure of the solvent

using Raoult's law $\Delta T_f = K' x_2$

x_2 mole fraction of solute

$$\Delta T_f = K' \frac{w_2/M_2}{w_1/M_1}$$

where, w_2 and w_1 are the masses and M_2 and M_1 are the molecular masses of the solute and solvent respectively

$$\Delta T_f = K' \frac{M_1}{1000} \cdot \frac{\frac{w_2}{M_2}}{w_1} \times 1000$$

$$= K_f \frac{w_2}{M_2 w_1} \times 1000$$

$$= K_f m$$

Exp-02

(a) what is activated charcoal?

when charcoal is heated in an inert gas at 625 - 1275 K, hydrocarbon and other gases are removed and then surface area is increases. This is more effective and known as activated charcoal.

(b) Define adsorption isotherm. write Langmuir and Freundlich adsorption isotherm equation and show their validity by drawing plots.

Adsorption isotherm:

If the temperature is kept constant the curve between a and p is called adsorption isotherm

$a = f(P)$ T is constant

a is the ~~adsorb~~ amount of adsorbed

p is the pressure.

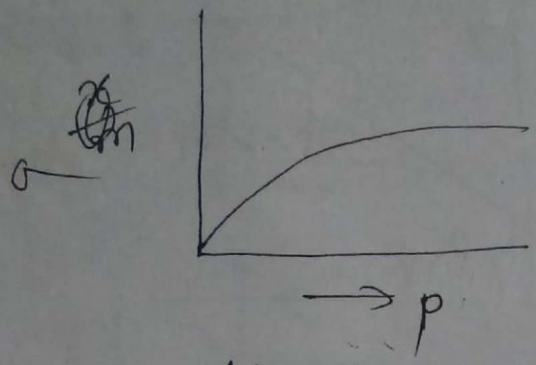


Fig: Adsorption isotherm

Langmuir adsorption isotherm: The Langmuir isotherm is represented by the following equation

$$\frac{x}{m} = \frac{K_1 K_2 c}{1 + K_1 c}$$

where K_1 and K_2 are constant. The above equation can be written as

$$\frac{c}{x/m} = \frac{1}{K_1 K_2} + \frac{c}{K_2}$$

Thus a plot of $\frac{c}{x/m}$ against c gives a straight line with a slope of $\frac{1}{K_2}$ and intercept of $\frac{1}{K_1 K_2}$ and shows the validity of Langmuir's adsorption isotherm.

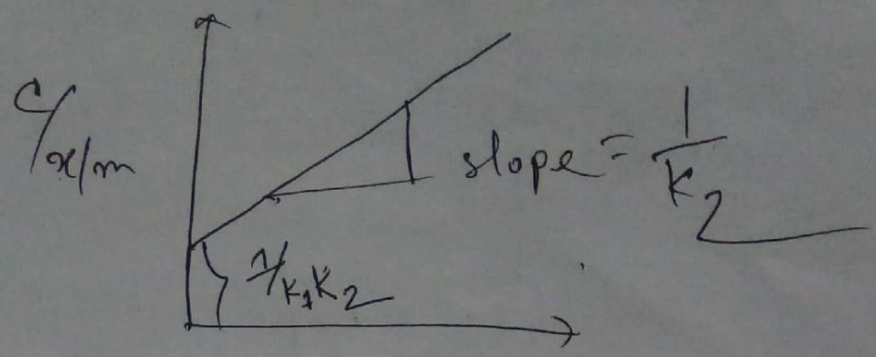


Fig: Langmuir adsorption isotherm

Freundlich adsorption isotherm.

Freundlich adsorption isotherm is represented by

$$x/m = Kc^{1/n} \quad \text{——— (1)}$$

x is the amount of solute adsorbed m, adsorbing material, c is the equilibrium concentration of adsorbate in solution n is another constant. 1/n is less than unity. On taking logarithm of eq (1)

$$\log \frac{x}{m} = \log K + \frac{1}{n} \log c$$

If a plot of $\log x/m$ against $\log c$ gives a straight line with a

slope of $\frac{1}{n}$ and intercept $\log K$ Freundlich adsorption isotherm is valid.

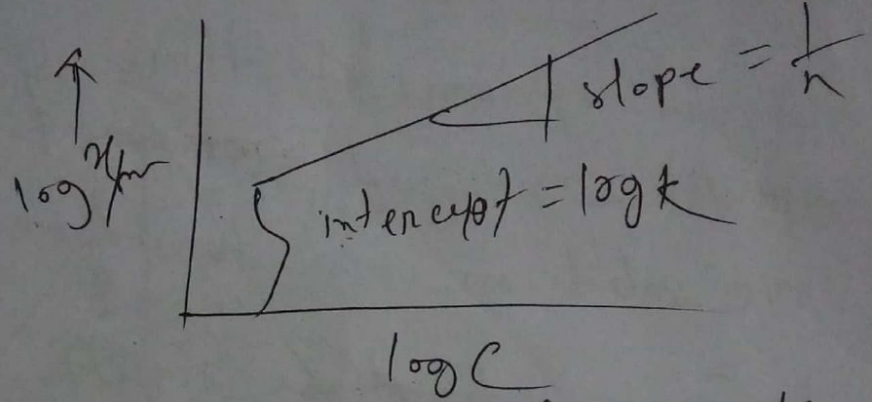


Fig. Freundlich adsorption isotherm.

Exp-03

① calculate the degree of freedom for three component system

Liquid systems of three components consis of three liquid components two liquids and one solid on one liquid and two solid

According to phase rule

$$F = C - P + 2 \quad \text{—— ①}$$

Here, F = degree of freedom

C = no. of component

P = no. of phase

For homogeneous system P = 1

we get from equation — ①

$$F = 3 - 1 + 2 = 4$$

For heterogeneous system P = 2

From eqn ① $F = 3 - 2 + 2 = 3$.

F = C - P + 2

Draw the phase diagram of three component system and show the homogeneous and heterogeneous area in the diagram.

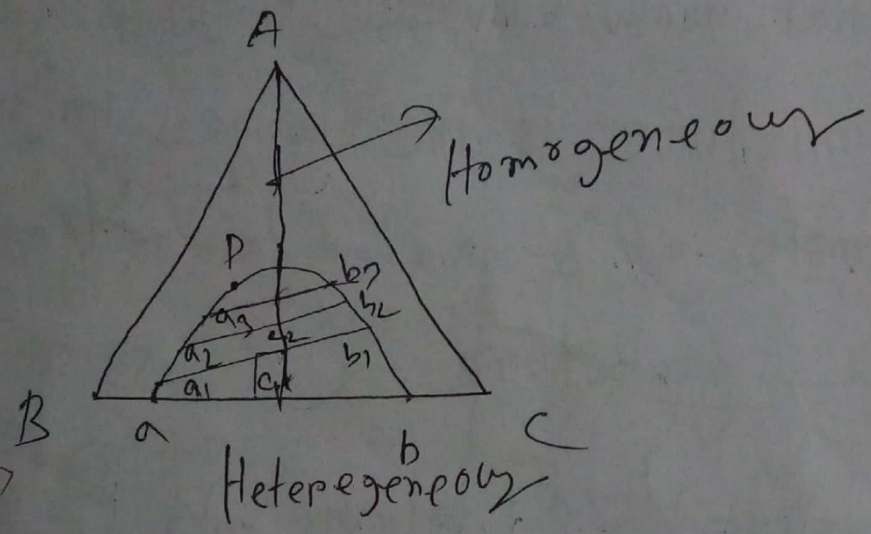


Fig: Three component system phase diagrame

Exp- 4, 5

Define partial molar volume?

partial molal volume of a substance A in a mixture is the change in volume per mole of A added to a large volume of the mixture

The partial molal volume V_j of a substance at same general composition is defined formally as follows

$$V_j = \left(\frac{\delta V}{\delta n_j}\right)_{P,T,n'}$$

where the susbscript n' signifies that the amount of all other substances present are constant.

Calculate the value of V_m for the mixture of A and B having the following parameter: Mole fraction of A = 0.1, M.W of A = 60, MW of B = 80, Density the mixture = 0.957 g/cm³.

$$V_m = x_A V_A + x_B V_B$$

$$= 0.1 \times 62.69 + 0.9 \times 83.59$$

$$= 6.269 + 75.2$$

$$= 81.469 \text{ cm}^3/\text{mol}$$

(Ans)

$x_A = 0.1$

or, $x_B = 1 - 0.1 = 0.9$

$$\rho = \frac{M}{V}$$

$$\Rightarrow V_A = \frac{M_A}{\rho} = \frac{60}{0.957}$$

$$= 62.69 \text{ g}$$

$$\Rightarrow V_B = \frac{M_B}{\rho} = \frac{80}{0.957}$$

$$= 83.59 \text{ g}$$

Exp- 07

Draw plots showing validity of Beer-Lambert law & write the equation of straight line for the plot. How could you calculate unknown concentration from the plot?

we know the Beer-Lambert law

$$A = \varepsilon C l \quad \text{---} \quad (1)$$

where A = Absorbance

C = concentration

l = thickness of medium

ploting A vs against C we get a straight line and that is

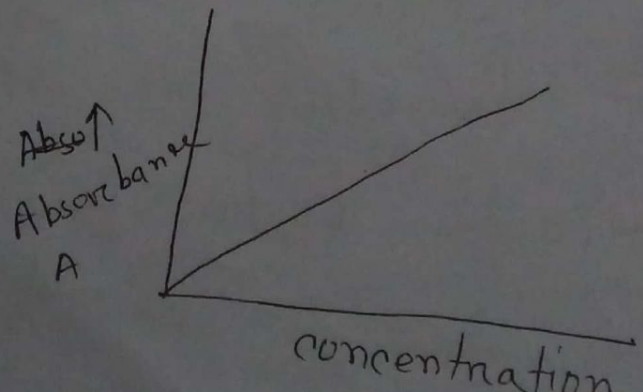


Fig: Beer's Law.

validity of Beer- Lambert Law

① there is no interaction between the solute of the solution

② obtained straight line joining through the origin, therefore this is the validity of Beer Lambert Law.

The straight line equation from the plot is

$y = mx$ where m = slope

By ~~putting~~ plotting A Vs c, we get a straight line passing through the origin

From the plot we can calculate the unknown strength of the solution.

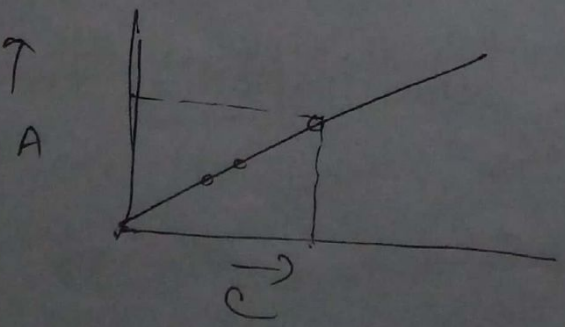


Fig: unknown conc.

Exp-08

write the reaction showing hydrolysis of the ester you have use in the experiment

In the presence of alkali ethyl acetate undergoes hydrolysis which is given below

$$CH_3COOC_2H_5 + \bar{O}H \rightleftharpoons CH_3COO^- + C_2H_5OH$$

The reaction follow second order rate law.

Describe how could you determine rate constant from conductance measurement.

rate law, $$K = \frac{1}{at} \cdot \frac{x}{(a-x)} \quad \text{---} \quad (1)$$

where a is the initial concentration in mol/l

K is rate constant

Now, C_0 and C_t are concentration at times zero and t

$$a \propto C_0 - C_\infty$$

$$x \propto C_0 - C_t$$

that means $a - x = C_t - C_\infty$

we get from eqn (1) $K = \frac{1}{at} \cdot \frac{C_0 - C_t}{C_t - C_\infty}$

$$C_t - C_\infty = \frac{1}{aK} \cdot \frac{C_0 - C_t}{t}$$

$$\Rightarrow C_t = \frac{1}{aK} \cdot \frac{C_0 - C_t}{t} + C_\infty \quad \text{---(2)}$$

Plotting C_t against $\frac{C_0 - C_t}{t}$ and we get a straight line

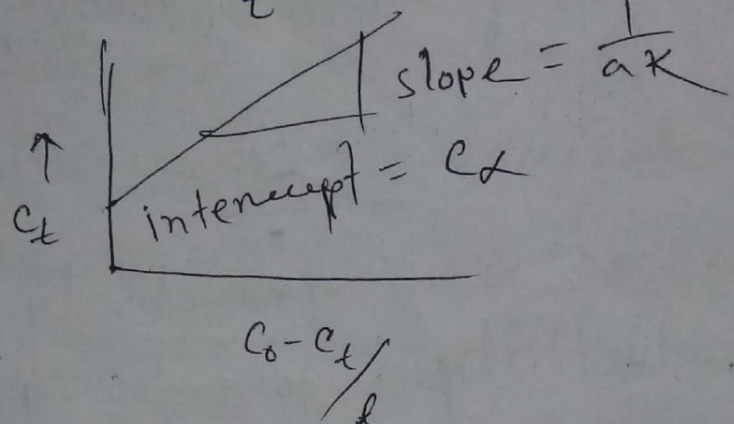


Fig: conductance measurement

unit of second order reaction rate constant $L\,mol^{-1}sec^{-1}$

what is hydrolysis?

Hydrolysis is defined as the interaction between the ions of salt and the ions of H_2O to given free H^+ or OH^- ions in soln

pure H_2O freely ionised into H^+ and OH^- ions.

$$H_2O \rightleftharpoons H^+ + OH^-$$

the salt AB will ionises as

$$AB \longrightarrow A^+ + B^-$$

The ions salt react with the ions of H_2O to give acidic or alkaline solution

$$A^+ + OH^- \rightarrow AOH$$
$$B^- + H^+ \longrightarrow BH$$

Exp-09

Define Molar conductivity Give units of molar conductivity

Molar conductance is defined as the conductance of all the ions produced ~~by~~ when 1 mole of an electrolyte is dissociated into it's ions in a volume V ml. It is ~~dete~~ denoted by Λ_m Mathematically

$$\Lambda_m = K \cdot V$$

where K specific conductance

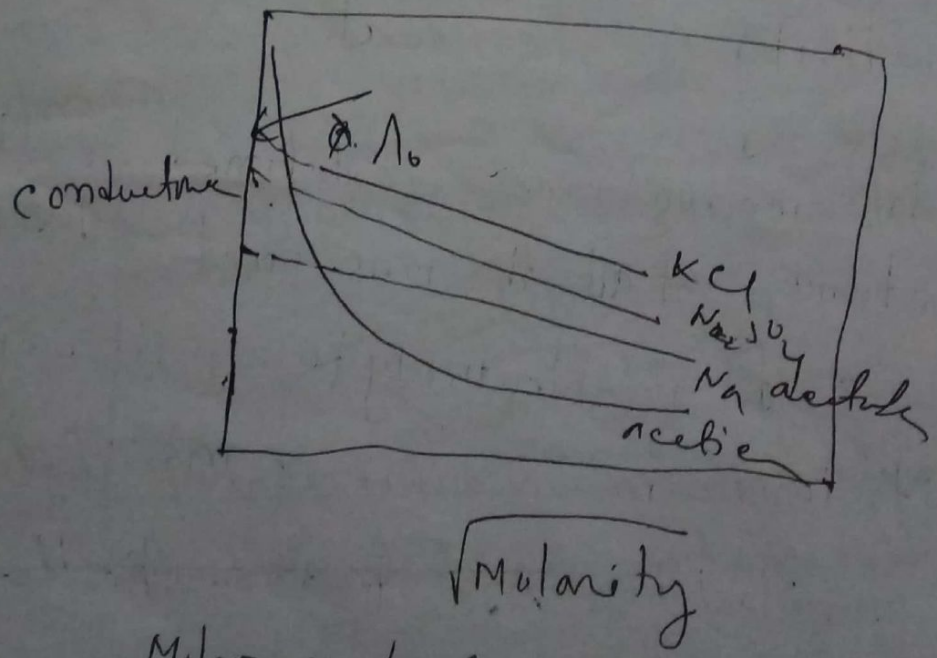
unit:

$$\Lambda_m = \frac{l}{RA} \cdot V \qquad = ohm^{-1} cm^2 mol^{-1}$$

$$= \frac{cm}{\Omega\, cm^2} \times \frac{cm^3}{mol}$$

why equivalent conductance increases with dilution

The valve of equivalent conductance increases on dilution. This increase is due to the fact that equivalent conductance is the product of specific conductance and the value of V of the solution containing 1 gm equivalent of the electrolyte.



Molar conductance Vs square root of concentration

what is degree of dissociation? write the value of K_c according to ostwald's dilution law.

The degree of dissociation is defined as the ratio of Λ_c to the Λ_0 and this is expressed mathematically $\alpha = \frac{\Lambda_c}{\Lambda_0}$

where, α = degree of dissociation

Λ_c = conductance of a concentration

& Λ_0 = conductance at zero concentration.

the value of K_c according to oswald dilution Law

$$K_c = \frac{\alpha^2 c}{1-\alpha}$$

Ostwald dilution is a relationship between the dissociation constant and the degree of dissociation of a weak electrolyte (acid base)

Draw and explain titration curve for the titration of acid by bases

① strong acid and strong bases

when the strong acid, HCl is titrated with strong base, NaOH, the hydrogen ions are removed by combination with the hydroxyl ions forming H_2O molecules and their place is take up by slow moving of Na^+ ions

$$H^+_{(aq)} + Cl^-_{(aq)} + Na^+_{(aq)} + OH^- \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O_{(l)}$$

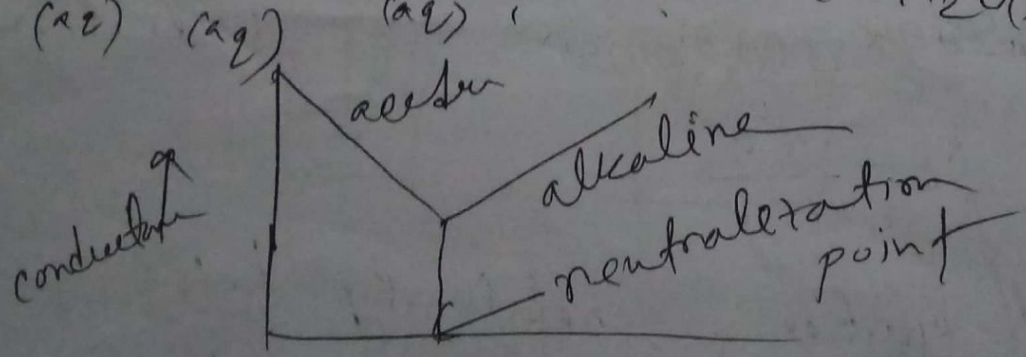
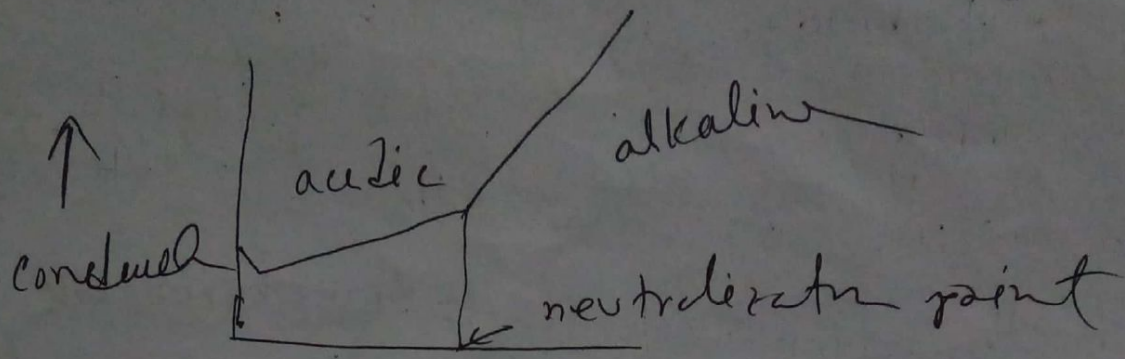


Fig: strong acid and strong base neutralization curve

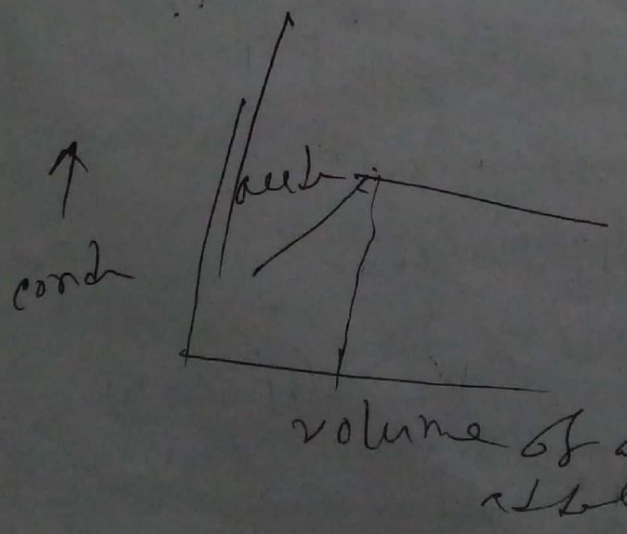
(ii) Titration of weak acid and strong base



$CH_3COOH + NaOH \longrightarrow$

(iii) weak acid and weak base

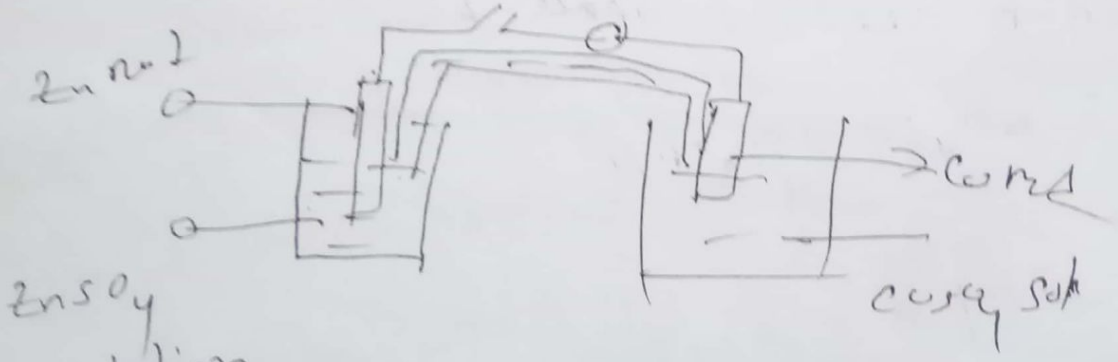
$NH_4OH + CH_3COOH$



what is salt bridge? why it is used

A salt bridge is a doubly bent tube containing a solution of KCl or, KNO_3 or, NH_4NO_3 and closed at the two ends by cotton wool so that when inverted the solution does not fall down.

The salt bridge is placed so that the liquid in the two half cell are connected



Daniel cell with a salt bridge

The salt bridge allows the follow of ions but prevents the mixing of the different solution.

(5) (a) A cell is prepared by

$Cu \rightarrow 0.34$

$\therefore Ni = -0.25$

(b) what will be the cell rxn

(b) calculate the standard EMF

(c) which electrode will be posn

(d) How will the cell is represented

(b) (c) anode: $Ni \rightarrow Ni^{2+} + 2e^-$ $\quad E^\circ_{Ni^{2+}/Ni} = -0.25$ V

cathode $\quad Cu^{2+} + 2e^- \rightarrow Cu$ $\quad E^\circ_{Cu^{2+}/Cu} = 0.34$ V

cell rotation:

$Ni(s) / Ni^{2+}(aq) \,||\, Cu^{2+}(aq) / Cu(s)$

(c) $E_{cell} = E_{cathode} - E_{anode}$

$= E_{Cu^{2+}/Cu} - E_{Ni^{2+}/Ni}$

$= 0.34 + 0.25$

$= 0.59$ volt (Ans)

(c) ~~Ni~~ / Ni electrode positive

(d)

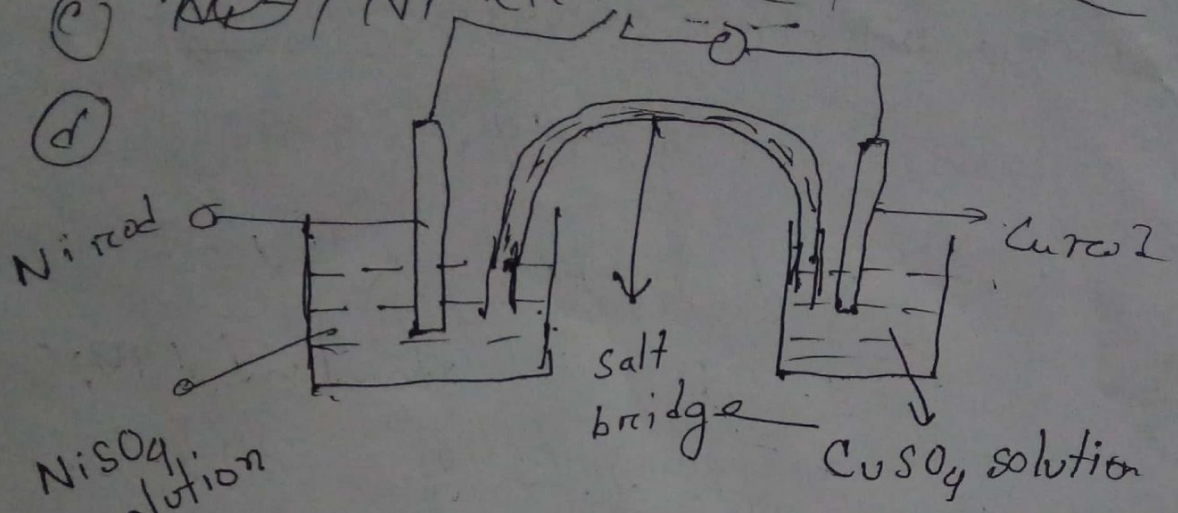


Figure: Electr

Ni^{2+} (aq) + 2e → Ni(s)

Half cell reaction for a cell Mg and Sn is as follows

$Mg \longrightarrow Mg^{2+}(0.01M) + 2e^-$ $\quad E^\circ = +2.34V$

$Sn^{2+}(0.1M) + 2e \rightarrow Sn$ $\quad E^\circ = -0.136$ V

(a) write the cell notation

(b) which electrode will be positive

(c) calculate the emf of the cell.

+2.33

(a) anode: $Mg \longrightarrow Mg^{2+} + 2e^-$

cathode: $Sn^{2+} + 2e^- \rightarrow Sn$

cell notation: $Mg \mid Mg^{2+}(0.01\ M) \,||\, Sn^{2+}(0.1M) \mid Sn$.

(b) Mg electrode is positive

(c) $E_{cell} = E_{cathode} - E_{anode}$

$E_{cathode} = E^\circ_{Sn^{2+}/Sn} + \frac{0.0591}{2} \log 0.1$

$= -0.136 + -0.029 = -0.165$ V